HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed and evaluated materials for new product development, ensuring compliance with stringent industry specifications and performance requirements.
- Conducted comprehensive material testing and analysis to assess durability, performance characteristics, and adherence to quality standards.
- Supported investigations into material failures and quality concerns, identifying root causes and proposing actionable corrective measures to manufacturing teams.
- Provided material-specific technical input and recommendations to product development teams, contributing to the successful launch of new product lines.
- Collaborated with manufacturing and quality assurance teams to enhance product performance and optimize production efficiency through material selection.
- Prepared detailed reports and presentations on test results and material analysis, communicating findings and recommendations to project stakeholders.
- Assisted in the evaluation of suppliers and the qualification process for new materials and components, ensuring supply chain integrity.

SKILLS

Materials Science – Material Selection, Material Testing, Material Analysis, Quality Assurance, Product Development, Failure Analysis, Root Cause Analysis, Supplier Evaluation, Material Compliance

Engineering Principles – Product Performance, Production Efficiency, Continuous Improvement, Sustainability, Cost Reduction, Industry Standards, Regulatory Compliance, Health and Safety

Data Analysis & Reporting – Test Result Analysis, Data Analysis, Report Preparation, Presentation Skills, Stakeholder Communication

Software & Tools – Jira, Confluence, Splunk, Dynatrace, CloudWatch, Python, Java, SQL, AWS, Docker, Jenkins, Git, Linux

Domain Knowledge – Glass-Reinforced Plastics (GRP), Composite Materials, Manufacturing Processes, Materials Technology

PROJECTS

GRP Composite Material Analysis and Testing (HCL Technologies)

- Performed detailed analysis on various Glass-Reinforced Plastics (GRP) composite materials to assess their mechanical properties and long-term durability for industrial applications.
- Developed and executed a standardized testing protocol for GRP samples, measuring tensile strength, flexural modulus, and impact resistance, identifying materials suitable for a key client's infrastructure project.
- Documented findings in comprehensive reports, including material specifications and performance data, which informed the final material selection for a high-volume production line.

Production Process Material Optimization (HCL Technologies)

- Investigated material performance within existing manufacturing workflows, identifying areas for improvement in terms of cost and production speed.
- Collaborated with the operations team to pilot alternative material formulations, analyzing their impact on product quality and manufacturing cycle times.
- Provided data-driven recommendations for material substitutions that led to a projected reduction in material waste by approximately 15%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)